

BEGINNER GUIDE & OUTPUTS

Linux System Resource Monitoring Tool

Step-by-Step Implementation Guide & Important Verification Outputs

ENVIRONMENT & REPOSITORY METADATA

- **Kernel:** WSL2 Linux 6.18.40.1-microsoft
- **Proc Entry:** /proc/resource_monitor
- **Languages:** C++17, C, Makefile
- **Git Branch:** device-driver (Clean)

Part 1: How to Implement the Project (Step-by-Step)

Step 1 — Setup Build Environment

Install compilers, ncurses terminal library, and kernel build headers:

```
sudo apt update && sudo apt install -y build-essential libncurses5-dev linux-headers-$(uname -r)
```

Step 2 — Project Architecture & Files

```
LinuxSystemMonitor/  
├─ include/      # C++ Headers (SystemMonitor.h, UI.h, ProcessManager.h)  
├─ src/          # C++ Source Files (SystemMonitor.cpp, UI.cpp, main.cpp)  
└─ driver/       # Kernel Module C Code (resource_monitor.c) & Makefile
```

Step 3 — Kernel Driver Implementation (driver/resource_monitor.c)

The Ring 0 C kernel module gathers metrics using kernel primitives:

si_meminfo(): Harvests physical RAM page counts (Total & Free RAM). **kcpustat_cpu():** Inspects per-CPU tick counters for CPU utilization. **for_each_process():** Traverses kernel task list to count Total, Running, and Sleeping processes. **proc_create():** Creates virtual interface file `/proc/resource_monitor`.

Step 4 — Build & Load Kernel Driver

```
cd ~/LinuxSystemMonitor/driver  
make                # Compiles resource_monitor.ko module  
sudo insmod resource_monitor.ko # Loads driver into Linux Kernel
```

Step 5 — User-Space C++ Application & ncurses UI

SystemMonitor.cpp opens /proc/resource_monitor via file stream, passes metrics to ProcessManager.cpp for CPU/Memory sorting, and renders in UI.cpp.

```
cd ~/LinuxSystemMonitor
make clean && make          # Compiles C++ monitor binary
./monitor                  # Launches interactive monitoring dashboard
```

Part 2: Important System Outputs to Show

Output 1: Kernel Driver Loaded

```
$ lsmod | grep resource_monitor
resource_monitor 12288 0
```

Output 2: Direct Proc Read

```
$ cat /proc/resource_monitor
Total RAM: 7823496 KB
Free RAM: 7020676 KB
```

Output 3: Standalone C++ Driver Test Output (driver/driver_test)

```
priya@Priya: ~/resource-monitor
Linux System Resource Monitor
Press Ctrl+C to quit

CPU usage:  0.0%
Memory used: 568.3 MB / 7640.6 MB (7.4%)

Top 5 processes by memory:
PID    NAME                MEMORY (MB)
264    unattended-upgr     31.8
162    networkd-dispat     29.0
48     systemd-journal     16.9
1      systemd             15.2
177    wsl-pro-service     15.1
```

Output 4: Main Interactive ncurses System Monitor Dashboard (./monitor)

```

                                LINUX SYSTEM MONITOR

CPU Usage   : 0.00% [OK]      Download : 0.00 MB/s
Memory Usage : 8.19% [OK]    Upload   : 0.00 MB/s
Disk Usage  : 6.48%          Processes: 26
Kernel Driver:               Sorted by: CPU
Linux System Monitor Kernel Module
=====

PID      PROCESS                CPU %      MEMORY MB
-----
223      unattended-upgr             0.00       31.86
6162     monitor                    0.00        5.11
5597     wsl-pro-service             0.00       15.06
407      bash                        0.00        5.41
381      (sd-pam)                    0.00        3.99
379      systemd                     0.00       13.31
337      login                       0.00        5.47
336      bash                        0.00        5.61
335      Relay(336)                  0.00        1.40
334      SessionLeader               0.00        1.27
250      chronyd                     0.00        2.34
232      chronyd                     0.00        7.70
230      agetty                      0.00        2.74
1        systemd                     0.00       15.05
200      rsyslogd                    0.00        5.85
180      systemd-logind              0.00        9.19
167      networkd-dispat             0.00       29.17
161      dbus-daemon                 0.00        5.32
160      cron                        0.00        3.10
159      chronyd-starter             0.00        1.96
89       systemd-udev                0.00       11.80
82       systemd-resolve             0.00        4.08
48       systemd-journal             0.00       18.93
8        init                        0.00        2.33
3        init-watcher                0.00        2.05
2        init-systemd(Ub)           0.00        2.28

[R] Refresh
[S] Sort CPU/Memory
[Q] Quit

```

INTERACTIVE KEYBOARD CONTROLS DEMO

- [R] Refresh telemetry and process table
- [S] Toggle process sorting between CPU % and Memory MB
- [Q] Quit application interface